Stockholm University

Department of Meteorology

Modelling of Large-Scale Circulation in Atmosphere and Ocean (MO8004)

Ocean Gyres (Munk)
Lab 4

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Questions

Theory

1. State the necessary assumptions that are needed to study the depth-integrated gyre circulation.
2. Write down the definition of the barotropic stream function.
3. Write down the equation for the stream function when assuming boundary condition $\Psi_s(x_e) = 0$. State the name of the circulation that is described by the equation.
4. Describe the shape of the gyre.
5. State the additional assumptions for the Munk theory.
6. Describe the main changes in the shape of the stream function under Munk's theory.
7. Give the equation for the Munk solution.
8. Describe the (shape of the) wind stress (discuss the equation for the wind stress) and the geographical situation where this can commonly be applied.

Experiments

Sverdrup circulation

1. Provide plots of the stream function for all your experiments at appropriate time steps.
2. What effect does $\beta = \partial f / \partial y$ have on the system?
3. What effect does the wind forcing have?
4. Provide cross sections (in x-direction) of the stream functions computed from your experiments.
5. From all your plots, discuss if the results resemble theory. Why / Why not?
6. Do the results improve if the size of the domain is increased?
7. What are the limitations of Sverdrup's model?

Munk's theory

1. Provide plots of the stream functions of your experiments.
2. Describe how the results have changed from Sverdrup circulation to Munk's theory.
3. Do theory and numerical solution correlate with each other?
4. Provide:
   - cross sections (in x-direction) of the stream functions computed from your experiments.
   - the difference of the cross sections of stream functions between theory and simulation results, and make sure that they also show the Munk thickness from theory.
5. From that, discuss the role of the friction coefficient for the shape.

1 Theory

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2 Experiments

2.1 Sverdrup circulation

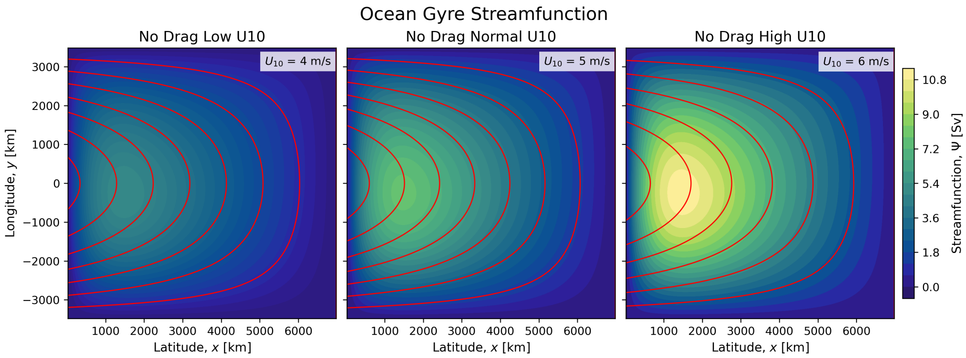

Figure 1: This figure...

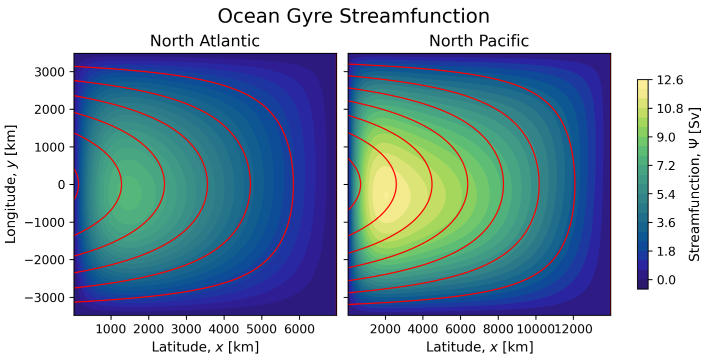

Figure 2: This figure...

2.2 Munks's theory

Note, we want

$$\frac{\delta_M}{\Delta x}$$

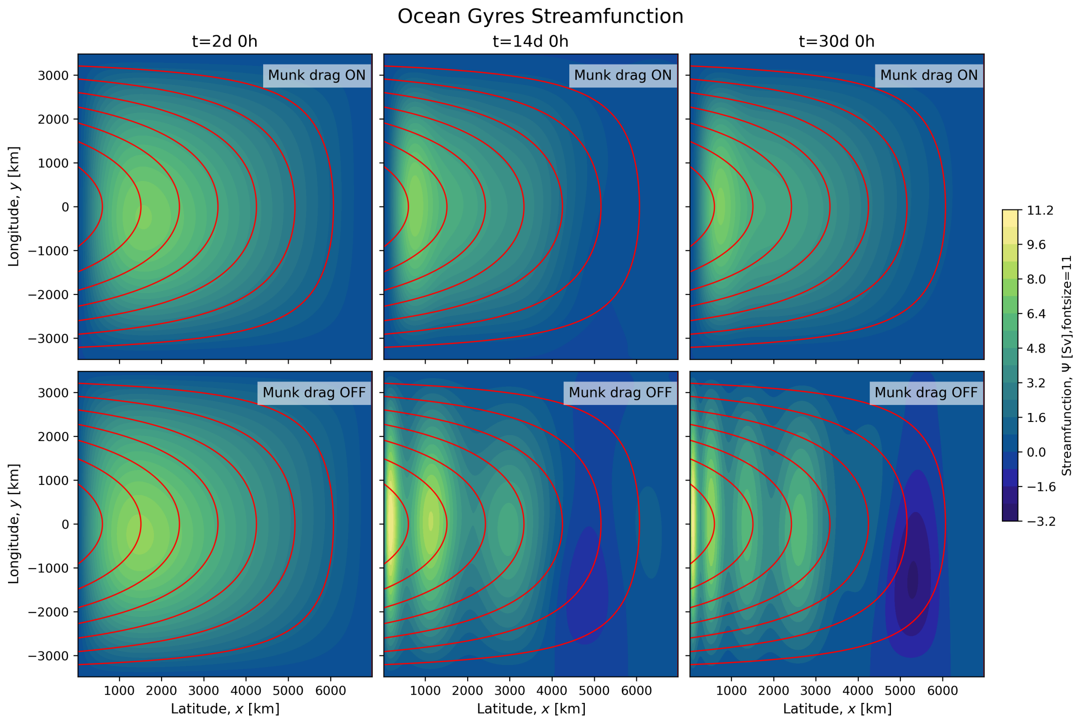

Figure 3: This figure...

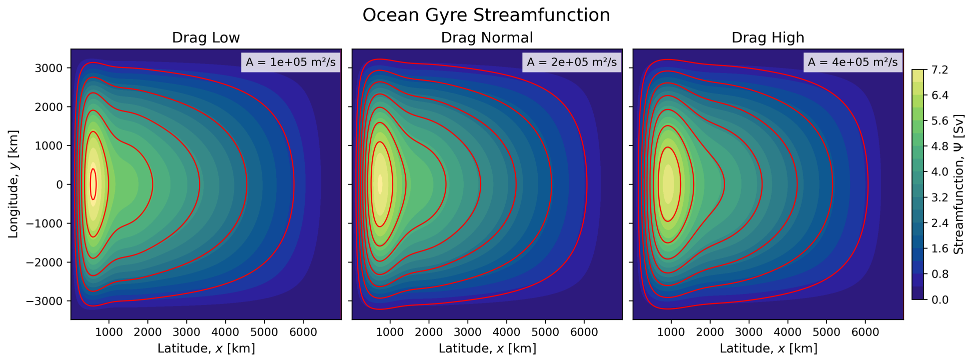

Figure 4: This figure...

A The Source Code

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/`
`Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`